An Evaluation of PQC Placement as Feature Extractor and Classifier in Hybrid Quantum CNN and ViT Models on CIFAR-10

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Abstract. Hybrid Quantum-Classical (HQC) models have been shown to offer a promising strategy for leveraging near-term quantum computing resources for machine learning tasks. While initial studies have shown promise on small datasets, it remains uncertain how HQC models would scale and what their architectures would look like for a more complex problem, such as CIFAR-10. This paper poses a fundamental question with regard to the emplacement of a Parameterized Quantum Circuit (PQC) in a deep learning architecture to ensure most effectiveness in a deep learning architecture. In this experiment, a comprehensive evaluation of HQC models using four different architectures has been done, where a PQC has been used either as a front-end feature extractor or a back-end classifier. Our findings show a striking point where models using a PQC as a classifier always outperform models using a PQC as a feature extractor, which are computationally expensive and create information bottlenecks. Notably, our proposed HQC-ViT model, which combines a classical ViT with a quantum classifier, outperforms a purely classical ViT model, proposing a valuable architectural blueprint for hybrid models under NISQ constraints. Ultimately, optimizing PQC placement in vision models provides important architectural guidelines for deploying resource-efficient, quantum-assisted diagnostic tools in future digital health and medical imaging applications.

Keywords: Hybrid Quantum-Classical Models · Image Classification · Parameterized Quantum Circuit (PQC) · Convolutional Neural Networks (CNN) · Vision Transformer (ViT) · CIFAR-10

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1 Introduction

Deep learning has transformed the field of computer vision by making it possible to do tasks like image classification and semantic segmentation better than ever before. Convolutional Neural Networks (CNNs) have been the most widely used architecture for many years. They use hierarchical layers of convolutions to learn spatial features well [1,2]. Even though these basic models are still very useful, the field is always changing, and more advanced architectures are showing better performance [3]. More recently, Vision Transformers (ViTs) have emerged as a powerful alternative, adapting the self-attention mechanisms that were transformative in natural language processing to vision tasks [4]. By treating an image as a sequence of patches, ViTs excel at capturing long-range, global dependencies, spawning a wide array of subsequent models and setting new state-of-the-art benchmarks [5].

At the same time, Quantum Machine Learning (QML) represents a new concept that has the potential to solve some computational challenges that can hardly be solved by classical computers [6]. Quantum circuits, by using quantum mechanical features like superposition and entanglement, can basically work in an exponentially large Hilbert space. This, in theory, opens the doors for developing models with drastically higher expressiveness. Most of the recent interest has focused on Variational Quantum Algorithms (VQAs), which aim to tackle complex problems by trying to find the best parameter setting for a quantum circuit [7].

However, the practical use of QML is presently limited by the restrictions of the quantum hardware. Currently, we are in the Noisy Intermediate-Scale Quantum (NISQ) era, which means the quantum devices are limited by the number of qubits, short coherence times, and high noise levels [8]. A pragmatic and hopeful way to proceed, therefore, is in Hybrid Quantum-Classical (HQC) models. These models establish a synergistic framework in which classical computers perform data processing and optimization, while a quantum co-processor is instructed to execute specific subroutines. The hybrid approach has been intensified as the main focus of the NISQ-era algorithm development. Thus, it represents the most feasible strategy to develop effective algorithms within NISQ constraints [9].

Rizvi et al. [10] introduced this hybrid strategy in computer vision by outlining two major architectures: the Hybrid Quantum-Classical Convolutional Neural Network (HQC-CNN) and the Hybrid Quantum-Classical Vision Transformer (HQC-ViT). Their work demonstrated the potential of these models on the MNIST handwritten digit dataset, showing competitive performance against classical counterparts.

While this foundational work is significant, its evaluation on MNIST leaves two critical questions unanswered. First, can these architectures scale effectively to more complex, multi-channel color image datasets? Second, and more fundamentally, where in a deep learning pipeline is a quantum component most effective? It is unclear whether a Parameterized Quantum Circuit (PQC) provides more benefit as a front-end feature extractor or as a back-end classifier.

Furthermore, CIFAR-10 will serve as a standardized benchmark for complex, multi-channel image classification, with the architectural insights provided being of tremendous relevance to digital health and medical imaging. The advancement of AI in healthcare largely relies on the development of efficient feature extraction and classification systems, which are capable of processing extremely complex health data with stringent computational constraints. The delineation of the most efficient placement of quantum components in a deep learning system will provide a fundamental architectural blueprint for future quantum-aided decision support systems. The translation of these hybrid systems from general computer vision benchmarks to specific health care tasks will enable the creation of extremely expressive, efficient systems for health care professionals.

This paper addresses both of these gaps. We extend the work of Rizvi et al. by migrating their architectures to the more challenging CIFAR-10 dataset. Furthermore, we introduce two novel "reversed" architectures, a CNN-QNN and a QNN-ViT, to conduct a systematic comparison of different quantum integration strategies. Our primary contributions are as follows:

- A systematic investigation of four distinct hybrid quantum-classical paradigms, comparing the use of a PQC as a front-end feature extractor versus a back-end classifier for both CNN and ViT backbones.
- A rigorous performance comparison of all four hybrid models against their equivalent, purely classical baselines to isolate and measure the impact of the quantum components.
- A comprehensive analysis of the trade-offs between model accuracy, the number of trainable parameters, and the computational cost of training for each paradigm.

This paper is organized as follows: In Sect. 2, we review related work in classical and quantum vision models. In Sect. 3, we describe our model architectures and experimental methodology. In Sect. 4, we present our findings. In Sect. 5, we offer a detailed discussion of these findings. In Sect. 6, we conclude our paper and propose future work.

2 Background and Related Work

This section provides an overview of the classical and quantum machine learning concepts that form the foundation of our work. First, we reviewed the core principles of Convolutional Neural Networks and Vision Transformers. Second, we introduced the necessary concepts from quantum computing for hybrid models, and situate our research within the existing literature on hybrid quantum-classical vision architectures.

2.1 Classical Vision Models

Convolutional Neural Networks (CNNs) represent a group of deep neural networks commonly used for image processing and analysis. The structural characteristics of CNNs are motivated by the human brain's visual cortex and have

been designed to learn spatial hierarchies of features in an automatic and adaptive way. The main unit of a CNN is the convolutional layer, which processes an input image using a set of filters to automatically and adaptively produce an output representation. These filters are small matrices that can represent basic patterns such as edges, colors, and textures. Higher layers combine basic patterns to automatically and adaptively detect more complex patterns, such as shapes and components of shapes or objects. Pooling layers are regularly placed between layers to gradually lessen the spatial dimensionality, which is instrumental in reducing the number of parameters and producing an invariant feature representation [1,2].

Vision Transformers (ViTs) denote a newer architectural model that has been a serious contender to the traditional CNNs' supremacy [4]. ViTs do not apply sliding-window convolutions to the images. Instead, the image is viewed as a series of non-overlapping, fixed-size patches. Each of these patches is then flattened and linearly projected into an embedding vector. To incorporate the spatial information, a learnable position embedding is added to each patch embedding. The series of vectors is then passed through a regular Transformer encoder, defined by a series of Multi-Head Self-Attention blocks. The self-attention mechanism allows the model to determine which other patches are most relevant to the one it represents, thereby helping the model to learn complex, long-range global dependencies within an image. This power is very effective, especially on large-scale datasets [5].

2.2 Quantum Machine Learning Concepts

Parameterized Quantum Circuits (PQCs) or Variational Quantum Circuits (VQCs) are essentially the quantum equivalent of a classical neural network layer. A PQC is defined as a quantum circuit consisting of a series of quantum gates, some of which are parameterized by classical variables, θ. In most cases, these gates would be single-qubit rotation gates ($R_X(\theta_i)$, $R_Y(\theta_j)$) alongside a fixed two-qubit entangling gate (CNOT). A Variational Quantum Algorithm aims to achieve this by training the parameters, θ, so that the quantum circuit represents the unitary transformation $\mathrm{U}(\theta)$ that solves the task at hand. Usually, the training is carried out by a classical optimizer that iteratively updates θ by utilizing the gradient of the cost function, which is derived from the circuit output [7].

Data encoding represents an essential procedure for transferring conventional data into a quantum state. As quantum algorithms work with quantum states, a mapping needs to be established to indicate how classical feature vectors are represented in the high-dimensional Hilbert space. One way, which is also the work here, is Angle Embedding, whereby every classical feature x_i is utilized to geometrically define the rotation of a gate, for instance, $R_X(\pi x_i)$. The selection of the data encoding method is the cornerstone of a QML model design, and it greatly determines the model's performance and expressive power [11].

Measurement is the process of measuring classical data from the quantum state following the execution of the circuit. One of the methods of measuring

classical data is to compute the expectation value of an observable for each qubit, like the Pauli-Z operator (σ_z), and then measure it. The result for each qubit is a classical number within the range $[-1, 1]$, which makes up the classical output vector and can be further processed by the next layers of a hybrid model.

2.3 Hybrid Quantum-Classical Vision Architectures

Hybrid architectural patterns have emerged as a result of the integration of PQCs into classical deep learning pipelines. Our paper is directed towards two of the most influential approaches in computer vision: quantum convolution and quantum classification.

Quantum Convolutional Layers: The concept of a Quantum Convolutional Neural Network (QCNN) basically means substituting a classical convolutional filter with a PQC. The fundamental ideas in the literature have been the first to demonstrate this concept by applying a small PQC to local patches of an input image, like a classical filter's receptive field [12,13]. The quantum circuit in this case, due to its potential to realize extremely complex and non-linear transformations, can be seen as a trainable feature extractor. The output pixels of the new feature map are formed by the measured results of the PQC, corresponding to each image patch. The original papers not only laid down the theoretical framework, but they were also mostly verified on toy or quantum-native datasets.

Building upon the foundational architectures proposed by Rizvi et al. [10], our research addresses their viability on the more complex, multi-channel CIFAR-10 dataset. Our primary contribution is a systematic investigation into a fundamental design question they left open: the optimal placement of the quantum component. We evaluate whether a PQC provides greater benefit as a front-end feature extractor or as a back-end classifier, providing a crucial analysis for designing effective hybrid models.

Quantum Classifiers: Another hybrid strategy is a robust classical CNN to extract features and a PQC as the final classification head. In such a setting, a classical CNN takes the input image and produces a compact, low-dimensional feature vector. This vector is subsequently used to encode a PQC that carries out the final classification task. The idea behind this is that the quantum circuit can embed classical features in a high-dimensional Hilbert space; thus, the quantum circuit can locate intricate correlations and decision boundaries in the data that are unattainable for a normal fully-connected layer [14]. They have been extensively studied for a myriad of applications, such as medical image analysis and transfer learning [15]. We have implemented this model as a focal reference to the quantum convolution model. This comparison enables us to locate the position of the deep learning pipeline where integration of a quantum component might bring the most advantage for CIFAR-10 classification.

2.4 Comparison to Prior Work

The landscape for CIFAR-10 is characterized by several key paradigms. Traditional methods, such as the specific CNNs examined in the study by Pant et

Table 1. Comparison of Related Work on CIFAR-10 and Positioning of Our Contribution.

Reference	Model Type	Specific Architecture	Key Finding/Focus on CIFAR-10	Gap Addressed by Our Work
Pant et al. [16]	Classical	Multiple CNN Architectures	Compared various classical CNN designs to identify the best-performing architecture specifically for CIFAR-10.	Provides a pure classical CNN baseline, but contains no quantum or Transformer components.
Tan [17]	Classical	Lightweight Vision Transformer (ViT)	Demonstrated that a lightweight ViT can outperform CNNs on CIFAR-10 without massive pre-training.	Establishes a strong, relevant classical ViT baseline, but does not explore hybrid quantum integration.
Henderson et al. [13]	Quantum Feature Extractor	Quanvolutional Neural Network (QNN)	introduced the "Quanvolutional" layer as a quantum feature extractor and demonstrated its potential on the MNIST dataset.	Represents the foundational proof-of-concept, but did not publish results on more complex datasets like CIFAR-10.
Trochun & Gordienko [18]	Hybrid Quantum	Quanvolutional NN & Hidden-Layer Replacement	Directly compared two different HQC designs on CIFAR-10, analyzing the trade-offs between them.	Investigated two hybrid paradigms, but did not include ViT-based models or quantum patch encoding.
This Work	**Hybrid Quantum**	**HQC-CNN, CNN-QNN, HQC-ViT, QNN-ViT**	**Systematic evaluation and comparison of multiple HQC designs on the full CIFAR-10 dataset.**	**Extends the Rizvi et al. models to CIFAR-10 and systematically compares four different HQC architectures (convolution, classifier, patch encoding) across both CNN and ViT backbones to identify the most effective integration strategy.**

al. [16] and the light Vision Transformer put forward by Tan [17], have thus formed a stable benchmark. At the same time, quantum and hybrid strategies have also been explored, from basic ideas such as "Quanvolutional" layers [13] to comparative studies of different hybrids implemented on CIFAR-10 datasets in [18].

Our work is primarily driven by the architectures proposed by Rizvi et al. [10]. They first explored the viability of HQC-CNN and HQC-ViT models on the MNIST dataset. However, it remains unclear whether they perform effectively on more complex real-world datasets, such as CIFAR-10, which includes color details and class complexity.

Thus, the current study makes a two-fold contribution to the literature: first, it generalizes the Rizvi et al. models to the more difficult CIFAR-10 dataset,

and second, it conducts a more comprehensive comparison than existing work by testing four different paradigms of HQC models based on two architectures: CNN and ViT. Table 1 clearly localizes this contribution within the current state-of-the-art and the specific gap filled in the current work.

3 Methodology

In this section, we define the experimental setup used to test various Hybrid Quantum-Classical (HQC) vision models on the CIFAR-10 dataset. We detail the data preprocessing steps and the specific neural network architectures for the four models under investigation. Additionally, this section outlines the design of the parameterized quantum circuits as well as the overall training.

3.1 Dataset and Preprocessing

In all experiments, the CIFAR-10 dataset [19] was used. It consists of 60,000 color images, each with a size of 32×32 pixels, and is evenly distributed among ten mutually exclusive categories, such as "airplane," "automobile," and "bird". The CIFAR-10 dataset is divided into a training set with 50,000 images and a test set with 10,000 images. CIFAR-10 serves as an excellent proxy for real-world visual complexity, such as the complex, high-dimensional patterns found in multi-modal health data and medical imaging, due to its high intra-class variance and multi-channel (RGB) format. This presents a significantly harder, yet more realistic, challenge than the simple grayscale MNIST dataset used in prior foundational works [10]. To ensure a fair comparison and robust model training, the following preprocessing and augmentation steps were applied:

1. Preprocessing for HQC-CNN: We needed to ensure compatibility with the architecture initially employed by Rizvi et al. [10] on 28×28 grayscale images from MNIST datasets. The preprocessing for this model, therefore, included converting all images to grayscale form. Then, the images were resized to 28×28 pixels.
2. Preprocessing for all other models: For the CNN-QNN and the two ViT models, the images were kept unchanged and remained with 3 channels (RGB colors). However, for the two ViT models, the image dimensions were changed from 32×32 to 28×28. This specific resizing was strictly necessary to allow the image dimensions to be evenly divided by the 16 non-overlapping 7×7 patches without altering the core transformer logic or limiting classical capacity. Additionally, all pixel intensities were scaled between 0 and 1.
3. Data Augmentation: To prevent overfitting and promote generalization, standard data augmentation methods were employed while training all models. The process included randomized horizontal flips, randomized rotations (less than 10°), and slight adjustments to the image width and height.

3.2 Model Architectures

We have implemented and tested four different HQC architectures along with classical baselines for comparison (where the quantum layer is replaced with a classical counterpart).

1. HQC-CNN (Quantum Convolutional Layer): This architecture, illustrated in Fig. 1, positions the quantum component at the beginning of the network. An input grayscale image is first processed by the QuantumConvLayer, which utilizes a 4-qubit parameterized quantum circuit (PQC) as a 2×2 convolution filter with a stride of 2. The resulting quantum feature map is then passed to a classical backend composed of two Conv2D blocks with batch normalization and dropout, followed by a GlobalAveragePooling2D layer and a final dense classifier.

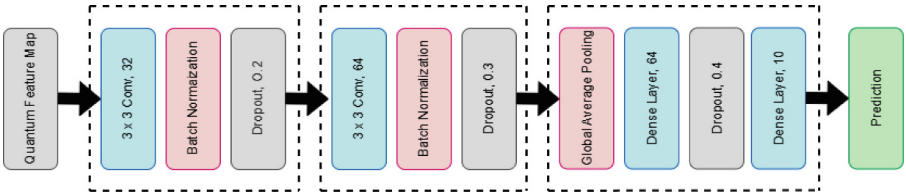


Fig. 1. The Hybrid Quantum-Classical CNN (HQC-CNN) Architecture. [Source: Rizvi et al., Mathematics, 2025], [10]

2. CNN-QNN (Quantum Classifier): In this model, illustrated in Fig. 2, a PQC is used as a back-end classifier. A classical CNN frontend, containing two convolution blocks (Conv2D + MaxPooling2D), first extracts spatial features from the 32×32 color images. The resulting feature map is flattened and passed through a dense layer, which reduces its dimensionality to match the number of qubits (4). This compact feature vector will then be encoded by the Quantum-Classifier Layer, and the outcomes of the parameterized quantum circuit will be used for a concluding dense layer for classification purposes with a softmax activation function.
3. HQC-ViT (Quantum Classifier Backend): This model, depicted in Fig. 3, couples a classical Vision Transformer (ViT) front-end with a PQC back-end. The

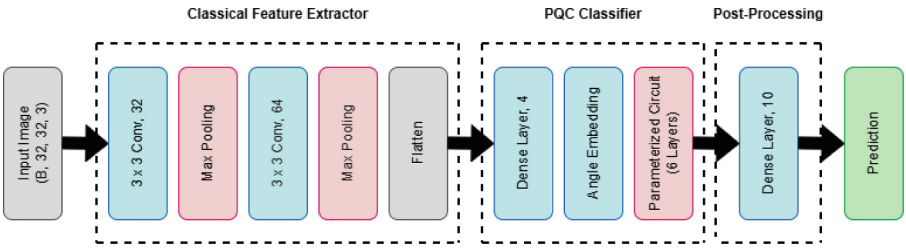


Fig. 2. The Hybrid CNN with a Quantum Classifier (CNN-QNN) Architecture.

ViT processes 28×28 color images by splitting them into 16 non-overlapping 7×7 patches and feeding them through a 4-layer Transformer encoder. The resulting [CLS] token embedding is projected into a 10-dimensional classical feature vector. This vector is then passed to a 10-qubit PQC to generate quantum features. Following the architecture of Rizvi et al. [10], a skip connection concatenates the original classical vector with the quantum feature vector, and a final linear layer performs classification on this combined result.

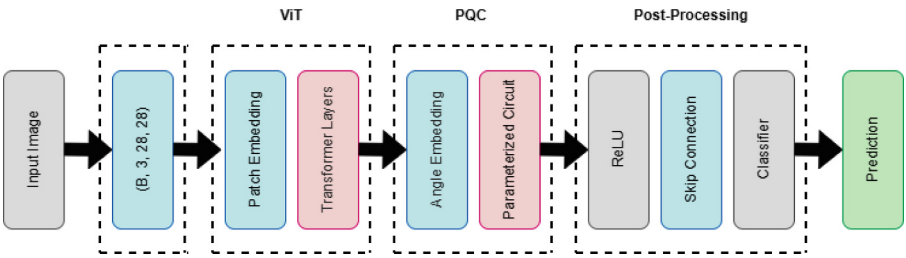


Fig. 3. The Hybrid Quantum-Classical Vision Transformer (HQC-ViT) Architecture. [Source: Rizvi et al., Mathematics, 2025], [10]

4. QNN-ViT (Quantum Patch Encoder): This final architecture, shown in Fig. 4, reverses the design of the HQC-ViT by using a PQC for feature extraction at the patch level. The input image is divided into 16 non-overlapping 7×7 patches. Each patch is then processed by a hybrid quantum-classical module: it is flattened, linearly projected to a 4-dimensional vector, and then encoded into a 4-qubit PQC. The measurement results from this quantum circuit are projected to the Transformer's embedding dimension (128), creating a sequence of quantum-enhanced patch embeddings. This sequence is then passed to a standard 4-layer Transformer backend for final classification based on the [CLS] token.

3.3 Quantum Circuit Design

The key building block for each hybrid model consists of a Parameterized Quantum Circuit (PQC). Despite variations in parameters, the common architecture includes data encoding and a trainable variational ansatz.

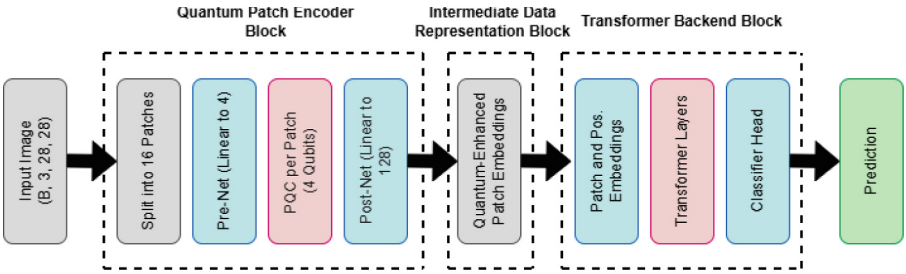


Fig. 4. The Quantum Patch Encoder Vision Transformer (QNN-ViT) Architecture.

- Data Encoding: For HQC-CNNs, qml.RX(np.pi*x_i, ...) gates are used to represent conventional values of each 2×2 patch. In other cases, qml.AngleEmbedding is used for encoding classical feature vectors.
- Variational Ansatz: The trainable part of the circuit aims to perform complex correlations. The HQC-CNN and CNN-QNN architectures use a series of single-qubit qml.RY gates followed by a string of entangling CNOT gates. This serves as their variational ansatz. The other models with a vision transformer architecture use qml.BasicEntanglerLayers for their variational ansatz. Details regarding qubit numbers for each architecture appear below in Table 2.

The number of quantum layers, or circuit depth, along with the precise count of trainable parameters for each PQC according to their use of variational ansatz (the single-qubit rotations and entangling gates) must be accurately tracked since they also contribute to the expressivity of the quantum components. The variational ansatz is repeated in the quantum circuit to form the full 'circuit depth'. For example, a CNN-QNN would have a circuit depth of 6 with respect to an execution on 4 qubits (resulting in 24 trainable parameters). On the other hand, the best performing HQC-ViT has a circuit depth of 5 on 10 qubits (resulting in 50 trainable parameters). This illustrates the parameter-efficient nature of quantum classifiers, in comparison to a classical dense layer (i.e., future comparisons can be drawn between the two methods). The complete architectural designs have been included in Table 2 below.

3.4 Experimental Setup

Both of the models were implemented using the Python programming language, which utilized the PennyLane library for the quantum components and TensorFlow, Keras, and PyTorch for the classical elements. The quantum circuits were run on the default qubit simulator. Training was done on a typical workstation computer, with the critical hyperparameters listed in Table 2. To enable a fair comparison and impose a time constraint on the training process, the EarlyStopping function was used with a patience of 10 epochs to monitor the validation loss. The best model was then selected for evaluation on the test set.

Table 2. Hyperparameter and Architectural Details.

Model	Framework	Optimizer	LR	Scheduler	Batch	Qubits	Q Layers	PQC Params
HQC-CNN	TF	Adam	1e-3	–	16	4	1	4
CNN-QNN	TF	Adam	1e-3	–	32	4	6	24
HQC-ViT	PyTorch	AdamW	3e-4	Cosine	64	10	5	50
QNN-ViT	PyTorch	AdamW	3e-4	Cosine	64	4	2	8

4 Results

This section provides the empirical results of the comparison study. In this study, the model was trained using the CIFAR-10 dataset until convergence. Early stopping on the validation loss was used to assess convergence. The three essential criteria used for model comparison are the test accuracy on the last epoch, the number of trainable parameters, and the average training time per epoch.

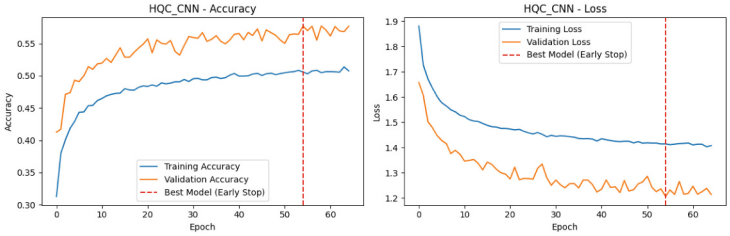
4.1 Overall Model Performance

The performance of the four hybrid schemes, along with their respective classical schemes, is given below in Table 3. There is a hierarchy of performance, where the approach of combining the quantum piece and the choice of the classical scheme are the controlling parameters.

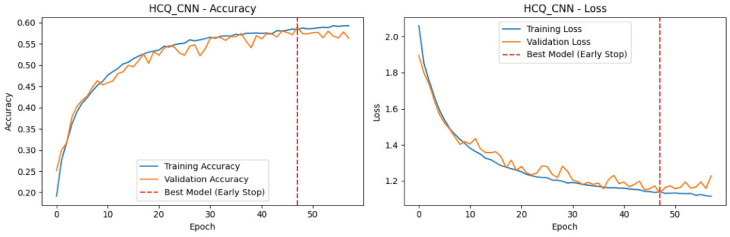
It is interesting to note that the architectures that make use of the quantum circuit as a back-end classifier perform better than those that use it as a front-end feature extractor. Additionally, the HQC-ViT model, which uses a Vision Transformer along with a quantum classifier, performed the best, registering a test accuracy of 67.68%, which is considerably better than the results of the CNN-based approaches.

4.2 Analysis of Convolutional Architectures

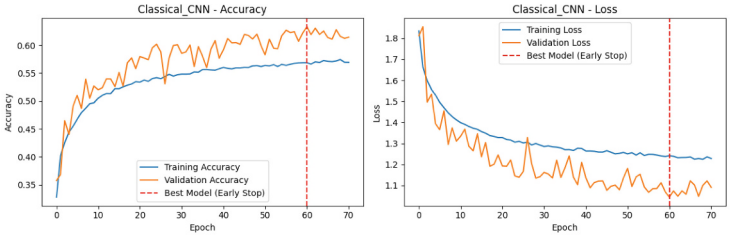
The two CNN models illustrate the essential trade-off between accuracy and computation intensity (Fig. 5). The HQC-CNN model setup, which used a PQC for the extraction of features on grayscale images, recorded a total accuracy of 57.39% but involved heavy computation, with the average time per epoch taking close to 90 min. This is because of the large overhead involved in the simulation of the quantum circuit for every 2×2 patch of images.



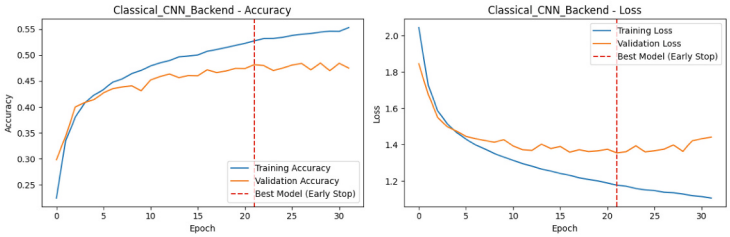
(a) HQC-CNN (Quantum Feature Extractor)



(b) CNN-QNN (Quantum Classifier)



(c) Classical CNN (Grayscale)



(d) Classical CNN (RGB))

Fig. 5. Training and validation dynamics for the convolutional architectures. The hybrid models (a, b) and their respective classical baselines (c, d) are shown.

By contrast, the CNN-QNN setup, where the PQC is used as the last classifier on the features extracted from the RGB images, reported a slightly better accuracy of 58.03%. Significantly, the training time of 48 s per epoch indicated that

Table 3. Overall Performance Comparison of HQC and Classical Models on the CIFAR-10 Test Set.

Model	Quantum Layer	Test Acc. (%)	Params (Trainable)	Train Time (s/epoch)
Convolutional Architectures (Grayscale 28×28)				
Classical-CNN	**No**	**62.38**	**24,702**	**≈34**
HQC-CNN	Yes (Feature Extractor)	57.39	24,686	≈5300
Convolutional Architectures (RGB 32×32)				
Classical-CNN	No	46.90	28,682	≈31
CNN-QNN	**Yes (Classifier)**	**58.03**	28,662	**≈48**
Transformer Architectures (RGB 28×28)				
Classical-ViT	No	63.68	552,568	≈189
HQC-ViT	**Yes (Classifier)**	**67.68**	**552,718**	**≈597**
QNN-ViT	Yes (Feature Extractor)	55.87	534,754	≈229

the HQC-CNN was surpassed by the CNN-QNN approach by over two orders of magnitude. Taken together, these findings make it clear that for CNN-based hybrid setups, the use of a quantum classifier at the tail end of the system is a much more viable and efficient approach.

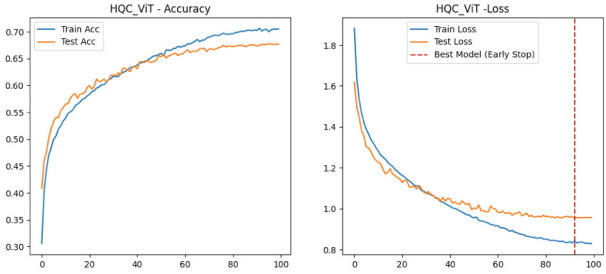
The substantial training time for HQC-CNN is due to the per-batch quantum circuit invocations required for every sliding window of the convolution. Expectation values were computed analytically via the statevector simulator.

4.3 Analysis of Vision Transformer Architectures

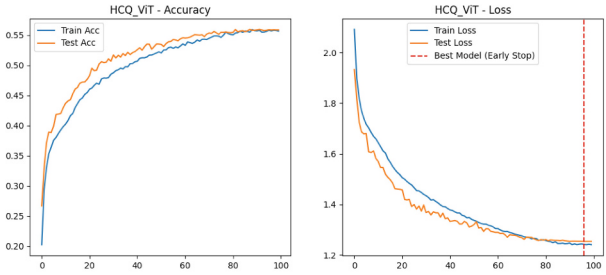
The results for ViT-based models had the highest level of variation, thus demonstrating the effects of quantum placement on the stronger classical baseline (Fig. 6).

The HQC-ViT model, where a classical ViT feature extractor is integrated with a 10-qubit quantum classifier, showed the best performance. It was observed that the model was able to achieve a test accuracy of 67.68%, outperforming the classical model's accuracy of 63.68%. This result suggests a potential quantum benefit, where replacing a linear classifier with a parameterized quantum circuit (PQC) not only does not harm but rather improves model performance despite a comparable number of parameters, as well as a slight increase in training time.

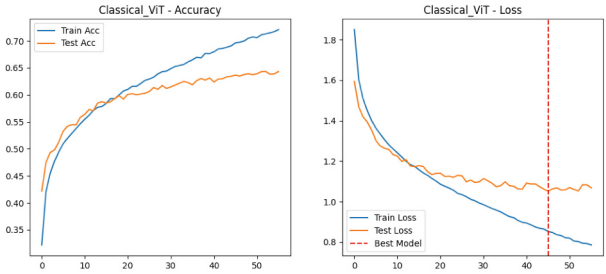
On the other hand, with a 4-qubit PQC to encode every image patch in the QNN-ViT model, the lowest performance was seen with a test accuracy of 55.87%. This shows that the information bottleneck caused by the limited qubits in the quantum circuit did not result in expressive embeddings for every patch, like those that were obtained in a classical linear projection. This result further supports that using a PQC as a front-end feature extractor for raw images in a complex classification problem like CIFAR-10 was suboptimal.



(a) HQC-ViT (Quantum Classifier)



(b) QNN-ViT (Quantum Feature Extractor)



(c) Classical ViT Baseline

Fig. 6. Training and validation dynamics for the Vision Transformer architectures. The hybrid models (a, b) and their classical baseline (c) are shown.

5 Discussion

This paper contributes to the construction of Hybrid Quantum-Classical models by investigating the best position of the Parameterized Quantum Circuit (PQC). Indeed, in this rapidly growing area, there are broadly speaking two paradigms considered at the present stage. The first is to consider the PQC as a front-end feature extractor inspired by ideas involving the notion of 'quanvolutional' layers [13], while the second involves the PQC as a back-end classifier for the

processed features in the classical domain [14,18]. The following experimental results confirm convincingly that the second approach strongly surpasses the first in the case of complex real-world datasets, such as CIFAR-10.

5.1 Key Findings Analysis

Our primary finding is that the architectural role of the Parameterized Quantum Circuit is the dominant factor for success in the models for the CNN as well as the ViT architecture.

Architectures featuring a PQC at the input stage, such as the HQC-CNN and the newly introduced QNN-ViT by the authors, are inherently not suited to such levels of model and/or data complexity. This assessment points to two significant problems that are probably inherent to such models. First, it would be prohibitively expensive to train such a model. Applying a PQC convolutionally, as in the HQC-CNN, resulted in a training time over 150 times longer than its classical counterpart. Moreover, these small PQCs act as a severe information bottleneck. This was demonstrated by the substantial degradation introduced in HQC-CNN's (57.39%) and QNN-ViT's (55.87%) performance compared to their baselines, indicating early "quanvolutional" methods' limitations for real-world-level vision tasks.

In the case of architectures employing a PQC as a back-end classifier, the effectiveness and feasibility become clear. This complementary combination, as incorporated in the CNN-QNN and HQC-ViT architectures, allows for the optimal performance of each component for its intended purpose. The classical part of the model is efficient in its capability to encapsulate high-dimensional images into a short, highly semantic feature vector, which is in turn appropriately processed by the PQC as a highly expressive classifier. This combination is computationally viable, incurring just a small overhead in terms of training time over the classical benchmarks. Much more significantly, it highlights the superior mathematical expressivity of quantum classifiers. The HQC-ViT model recorded 67.68%, which is an improvement over its robust classical baseline of 63.68%. Notably, the CNN-QNN model recorded 58.03%, which is an 11-point improvement over its classical model's 46.90% performance. We hypothesize that the presence of the PQC, projecting classical features into a high-dimensional Hilbert space, can identify intricate correlations and decision boundaries that are inaccessible to a simple classical dense layer.

5.2 Current Challenges and Limitations

The results presented in this paper provide architectural insights that need to be contextualized within the current quantum computing paradigm. It is noteworthy that the experiments carried out in this research were conducted using classical simulators to provide a clean environment without errors. As such, the performance improvements reported in this paper are purely a result of increased mathematical expressiveness rather than actual quantum speedup. The practical realization of the architectural models presented in this research would be

heavily influenced by various hardware-related aspects of the NISQ processors. Furthermore, the number of qubits utilized in the proposed architectural models is relatively low (between 4 and 10), implying that the full expressiveness of these architectures may not be realized until more sophisticated quantum computing architectures are developed.

In addition to this, due to the computational complexity of quantum convolutions carried out using classical simulators (which takes approximately 5300 s per epoch for the HQC-CNN), the evaluation of the proposed architectural models is carried out using fixed random seeds for reproducibility. Future work will focus on statistical validation of the results reported in this research, including the computation of the mean and standard deviation for multiple runs of the experiment.

It is noteworthy that the quantum circuits and data encoding schemes utilized in this research are relatively simple. Future work will focus on exploring more complex ansatz circuits to more efficiently map classical information onto quantum bits. A potentially valuable extension of the work presented in this research is to utilize the successful realization of the HQC-ViT architecture for more complex tasks related to medical imaging and digital health.

5.3 Applications to Digital Health and Medical Imaging

One of the most important outcomes of this benchmarking study is the direct applicability to the future of digital health and medical image analysis, which is in line with the growing need to develop sophisticated, data-driven decision support systems in healthcare settings. In fact, machine learning problems in medical image analysis, like histopathological image analysis, MRI analysis, and anomaly detection in radiographs, are extremely challenging due to the high dimensionality of the data and the complex correlations needed to make accurate diagnoses.

The key outcome of this study, which is the marked effectiveness and computational feasibility of Parameterized Quantum Circuits (PQCs) when used as back-end classifiers, rather than front-end feature extractors, offers a practical roadmap to the incorporation of Quantum Machine Learning in AI in healthcare settings. In fact, in medical image analysis, front-end quantum convolutions, or "quanvolutional" layers, have been shown to have computationally intractable costs when processing large, high-resolution medical images in a healthcare setting. In this regard, the Hybrid Quantum-Classical Vision Transformer (HQC-ViT) framework, which has been shown in this study, is the optimal solution to this problem. In this workflow, a classical neural network is used to map high-resolution medical images into dense, semantic feature representations, which are subsequently encoded by the PQC, which is used exclusively in the classification stage. The PQC maps patient features to a high-dimensional Hilbert space, where complex, non-linear decision boundaries may be identified, which may enable the distinction between malignant and benign disease.

Accordingly, the architectural guidelines outlined in this study provide a foundation upon which future, next-generation diagnostic tools may be devel-

oped, with future work building upon this foundation by incorporating the optimized HQC-ViT and CNN-QNN architectures on specific medical image datasets, like MedMNIST or ISIC melanoma datasets, to validate the utility of quantum classifiers in providing sophisticated, model-based decision support to medical professionals.

6 Conclusion

In our research, we conduct a comprehensive analysis of the optimal integration approaches for Parameterized Quantum Circuits (PQCs) in both Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) for CIFAR-10 classification. Through our experiment of four different hybrid approaches with their classical counterparts, we aim to find where in a deep learning pipeline a quantum part provides superior improvement.

The above analysis provides a distinct and conclusive finding: using a PQC as a back-end classifier clearly outperforms it as a front-end feature extractor in terms of both performance and computational efficiency. Front-end quantum layers, when directly used on image patches, are severe information bottlenecks and are computationally utterly impractical for this size of data, with consistent subpar performance compared to their classical counterparts.

On the other hand, HQC-ViT, which uses a classical ViT for feature extraction and a PQC for classification, not only remains computationally feasible but also displays the highest test accuracy among all models. It shows a subtle yet noticeable performance improvement over the classical model, which demonstrates that replacing a classical classifier with a PQC can improve expressivity and model performance, offering valuable architectural guidance for hybrid design in the NISQ era.

Our results offer a strong recommendation for researchers and practitioners in the NISQ era. The most promising and practical approach for hybrid quantum-classical vision is to focus on architectures where PQCs are used as powerful, expressive classifiers for classically processed data. Future work should therefore investigate more advanced data encoding strategies and benchmark these architectures on real quantum hardware to evaluate their performance under realistic noise conditions. Furthermore, applying the HQC-ViT paradigm to other challenging domains like medical imaging and exploring co-design strategies for hardware-efficient quantum circuits are valuable next steps.

Applying this HQC-ViT paradigm to challenging healthcare domains, such as medical imaging and patient data analysis, represents a highly promising direction for future quantum-enhanced decision support systems.

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